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MID TERM EXAMINATION
SECOND SEMESTER [B.TECH] JUNE 2022

Paper Code: BS-106

Subject: Applied Physics-II

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt three questions in all. Q. No. 1 is compulsory and attempt any two questions from the remaining.

Q. No.	Question	Marks
Q1	Attempt all of the following:	(5×2=10)
(a)	Which of these is an acceptable wave function: $\Psi = 1/x$ or $\Psi = e^{-x^2}$? Give reason.	(2)
(b)	Calculate the de-Broglie wavelength of a 4 keV electron.	(2)
(c)	Show that the energy difference between the n^{th} and $(n + 1)^{\text{th}}$ eigenstate of a particle in an infinite well (1D) is proportional to (2n + 1) .	(2)
(d)	Categorize the following particles as Bosons and Fermions. Give reasons too: (i) Photon (ii) Electron (iii) α -particles (iv) Hydrogen Nucleus	(2)
(e)	An electron gas obeys MB statistics. Calculate the average thermal energy (in eV) of an electron gas system at 300 K.	(2)
Q2 (a)	Set up Schrödinger equation for a particle in an infinite well (1D). Solve it for eigen wave function and energy.	(5.5)
(b)	Using Heisenberg's uncertainty principle, prove the non-existence of electrons in the nucleus.	(2.5)
(c)	Compute the transmission probability for an electron of energy 2.0 eV if barrier is 10.0 eV high and 1.0 nm wide.	(2)
Q3 (a)	Discuss the Black Body spectrum. Using Bose-Einstein statistics, derive an expression for Planck's radiation law.	(3 + 3.5)
(b)	The density of Zinc is $7.13 \times 10^3 \text{ kg/m}^3$ and its atomic weight is 65.4. Calculate the Fermi energy and mean energy at 0 K.	(3.5)
Q4 (a)	Compare the salient features of Maxwell-Boltzmann (MB), Bose-Einstein (BE), and Fermi-Dirac (FD) statistics.	(5)
(b)	Compute the expectation value for position of a particle in a 1D infinite well.	(3)
(c)	Give a brief account of Quantum mechanical Tunnelling.	(2)

**** End of Question Paper ****